

## ***PREFORMULATION TESTING OF SOLID DOSAGE FORMS***

**Preformulation testing** is the first step in the rational development of dosage forms of a drug substance.

- ↳ *It can be defined as an investigation of physical and chemical properties of a drug substance - alone and when combined with excipients.*
- ↳ *The overall objective of **preformulation testing** is to generate information useful to the formulator in developing **stable** and **bioavailable** dosage forms which can be **mass-produced**.*

- ↳ During the early development of a new drug substance, the synthetic chemist, alone or in cooperation with specialists in other disciplines (including preformulation), may record some data which can be appropriately considered as preformulation data.

- ↳ **This early data collection may include such information as**

- gross particle size,
- melting point,
- infrared analysis,
- thin-layer chromatographic purity,
- and other such characterizations of different laboratory-scale batches.

- ↳ **These data are useful in guiding, and becoming part of, the main body of preformulation work.**

## Steps in Preformulation Process Pharmaceutical Research

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Stability</b><br><b>a. Solid State</b><br>(1) Temperature<br>(2) Light<br>(3) Humidity<br><b>b. Solution</b><br>(1) Solvent<br>(2) pH<br>(3) Light<br><b>2. Solid State Compatibility</b><br><b>a. TLC Analysis</b><br><b>b. DRS Analysis</b><br><b>3. Physico-chemical Properties</b><br><b>a. Molecular Structure and Weight</b><br><b>b. Color</b><br><b>c. Odor</b><br><b>d. Particle size, Shape, and Crystallinity</b><br><b>e. Melting Point</b><br><b>f. Thermal Analysis Profile</b><br>(1) DTA<br>(2) DSC<br>(3) TGA<br><b>g. Hygroscopicity Potential</b><br><b>h. Absorbance Spectra</b><br>(1) UV<br>(2) IR | <b>i. Solubility</b><br>(1) Water and Other Solvents<br>(2) pH Solubility Profile<br>(3) Salt Forms<br>(4) Cosolvents<br>(5) Complexation<br>(6) Prodrug<br><b>j. Effect of pH on UV Spectra</b><br><b>k. Ionization Constant</b><br><b>l. Volatility</b><br><b>m. Optical Activity</b><br><b>n. Polymorphism Potential</b><br><b>o. Solvate Formation</b><br><b>4. Physico-mechanical Properties</b><br><b>a. Bulk and Tapped Density</b><br><b>b. Compressibility</b><br><b>c. Photomicrograph</b><br><b>5. In Vitro Availability Properties</b><br><b>a. Dissolution of Drug Crystal Per se</b><br><b>b. Dissolution of Pure Drug Pellet</b><br><b>c. Dissolution Analysis of Pure Drug</b><br><b>d. Rat Everted Gut Technique</b><br><b>6. Other Studies</b><br><b>a. Plasma Protein Binding</b><br><b>b. Effect of Compatible Excipients on Dissolution</b><br><b>c. Kinetic Studies of Solution degradation</b><br><b>d. Use of Radio-labeled Drug</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

---

---

---

---

---

---

---

- ↳ The formal preformulation study should start at the point after **biological screening**, when a decision is made for further development of the compound in **clinical trials**.
- ↳ Before embarking upon a formal program, the preformulation scientist must consider the following:
- 1. The available physicochemical data (including chemical structure, different salts available)**
  - 2. The therapeutic class of the compound and anticipated dose**
  - 3. The supply situation and the development schedule (i.e., the time available)**
  - 4. The availability of a stability-indicating assay**
  - 5. The nature of the information the formulator should have or would like to have**

---

---

---

---

---

---

---

---

## 1. ORGANOLEPTIC PROPERTIES

### 1.1 Color

Unappealing to the eye ==> instrumental methods or variable from batch to batch

Record of early batches ==> establishing "specs" is very useful for later production

Undesirable or color ==> incorporation of a dye variable in the body or coating

---

---

---

---

---

---

---

---

## 1.2 Odor and Taste

Unpalatable ==> use of less soluble chemical form  
(bioavailability not compromised!)

==> suppressed by - flavors  
- excipients  
- coating

Drug substances  
irritating to skin  
or sternutatory  
(sneezing) ==> handling precautions

Flavors, dyes, excipients used ==> stability  
bioavailability

Table 1. Suggested Terminology to Describe  
Organoleptic Properties of Pharmaceutical Powders

| Color        | Odor      | Taste     |
|--------------|-----------|-----------|
| Off-white    | Pungent   | Acidic    |
| Cream yellow | Sulfurous | Bitter    |
| Tan          | Fruity    | Bland     |
| Shiny        | Aromatic  | intense   |
|              | Odorless  | Sweet     |
|              |           | Tasteless |

## 2. PURITY

- ⌘ Materials with impurities not necessary to be rejected
- ⌘ Another control parameter for comparison with subsequent batches
- ⌘ More direct concerns - impurity can affect:
  - Stability: metal contamination in ppm
  - Appearance: off-color -> recrystallized -> white
  - Toxic: aromatic amine (p-amino phenol) -> carcinogenic
- ⌘ Often remedial action => simple recrystallization

## DEGRADATION PRODUCTS

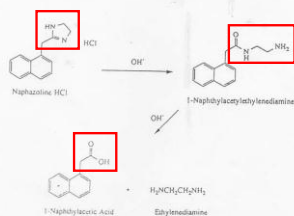
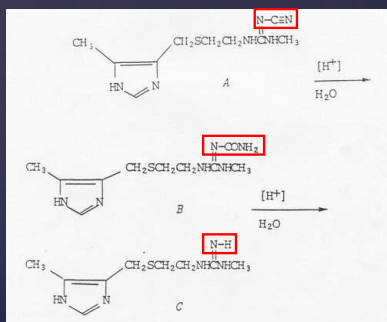


Figure 12. Degradation products of naphazoline hydrochloride under alkaline conditions.



## Cimetidine-acid hydrolysis

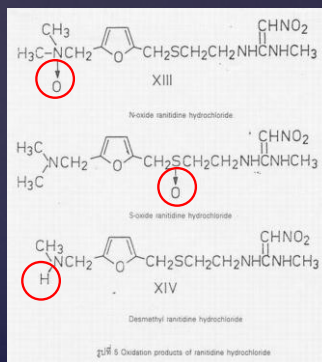
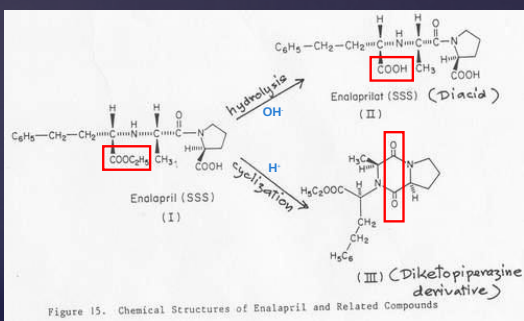
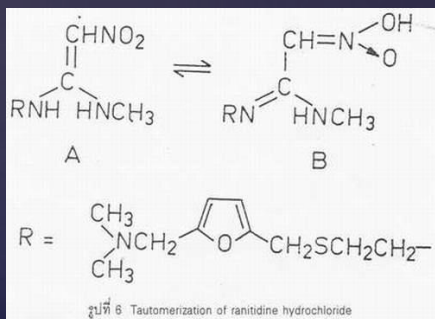


Figure 13. Oxidation products of ranitidine hydrochloride.



Techniques used for characterizing purity are the same as used in preformulation study :

- Thin layer chromatography (TLC)
- High-pressure liquid chromatography (HPLC)
- Gas chromatography (GC)

**Impurity index (II)** defined as the ratio of all responses (peak areas) due to components other than the main one to the total area response.

**Homogeneity index (HI)** defined as the ratio of the response (peak area) due to the main component to the total response.

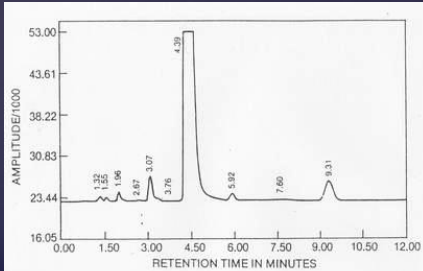


Figure 1 HPLC chromatograph of a typical batch of an experimental drug.

**Example:**

Main component - retention time: 4.39 min

- area response: 4620

Impurities - 7 minor peaks

- total area response : 251

$$\text{Impurity index} = 251 / (4620 + 251) \\ = 0.0515$$

$$\text{Homogeneity index} = 1 - 0.0515 \\ = 0.9485$$

↳ **USP Impurity Index** defined as a ratio of responses due to impurities to that response due to a defined concentration of a standard of the main component. (using TLC)

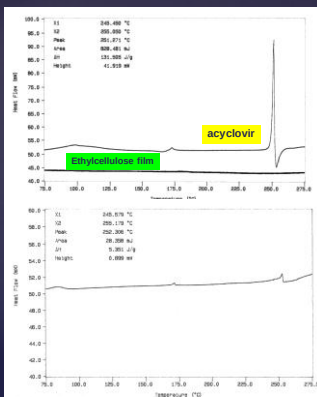
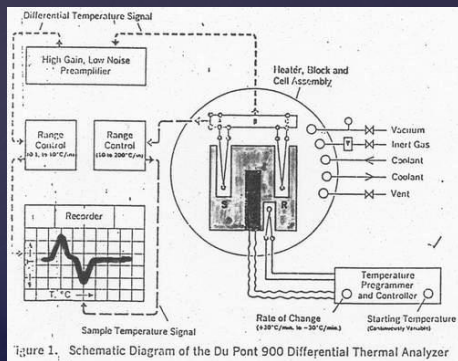
**General limit**     2 % impurities

↳ All **II**, **HI**, **USP II** are not absolute measures of impurity since the specific response (molecular absorbances or extinction coefficient) due to each impurity is assumed to be the same as that of the main component.

↳ **More accurate analysis** - identification of each individual impurity followed by preparation of standards for each one of them.

to: Other useful tools in assessment of impurity:

- Differential Thermal Analysis (DTA)
- Thermogravimetric Analysis (TGA)
- Differential Scanning Calorimetry (DSC)
- Powder X-Ray Diffraction (PXRD)

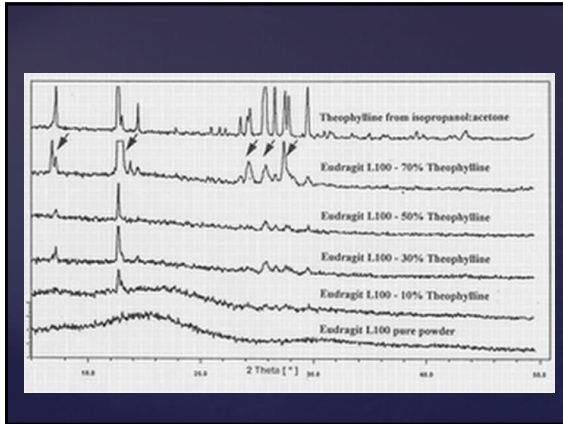


DSC thermograms of pure acyclovir and pure ethylcellulose films

DSC thermograms of ethylcellulose film containing 12.8 % acyclovir with 15 % propylene glycol and 10 % Tween 80







### 3. PARTICLE SIZE, SHAPE, AND SURFACE AREA

Effects of particle size distribution and shape on:

- Chemical and physical properties of drug substances.
- Bioavailability of drug substances (griseofulvin, chlorpropamide).
- Flow and mixing efficiency of powders and granules in making tablets.
- Fine materials tend to require more amount of granulating liquid (cimetidine).
- Stability, fine materials relatively more open to attack from atmospheric  $O_2$ , heat, light, humidity, and interacting excipients than coarse materials. (Table 2)

Table 2. Influence of Particle Size on Reaction of Sulfacetamide with Phthalic anhydride in 1:2 Molar Compacts after 3 hr at 95 °C

| Particle size of<br>sulfacetamide<br>( $\mu m$ ) | % Conversion<br>$\pm$ SD |
|--------------------------------------------------|--------------------------|
| 128                                              | 21.54 $\pm$ 2.74         |
| 164                                              | 19.43 $\pm$ 3.25         |
| 214                                              | 17.25 $\pm$ 2.88         |
| 302                                              | 15.69 $\pm$ 7.90         |
| 387                                              | 9.34 $\pm$ 4.41          |

Weng and Parrott

- Very fine materials are difficult to handle, overcome by creating solid solution in a carrier (water-soluble polymer).
- Important to decide, maintain, and control a desired size range.
- Safest - grind most new drugs with particle diameter > 100  $\mu\text{m}$  (~ 140 mesh) down to ~ 10 - 40  $\mu\text{m}$  (~ 325 mesh).
- Particles with diameter  $\leq 30 \mu\text{m}$  (~ 400 mesh), grinding is unnecessary except needle-like => improve flow.
- Drawbacks to grinding:
  - material losses
  - static charge build-up
  - aggregation => increase hydrophobicity  
=> lowering dissolution rate
  - polymorphic or chemical transformations

---

---

---

---

---

---

---

---

### 3.1 General Techniques For Determining Particle Size

#### 3.1.1 Microscopy

- Most rapid technique.
- But for quantitative size determination requires counting large number of particles.
- For size ~ 1  $\mu\text{m}$  upward (magnification x400).
- Suspending material in nondissolving fluid (water or mineral oil)
- Polarizing lens to observe birefringence => change in amorphous state after grinding?

---

---

---

---

---

---

---

---




---

---

---

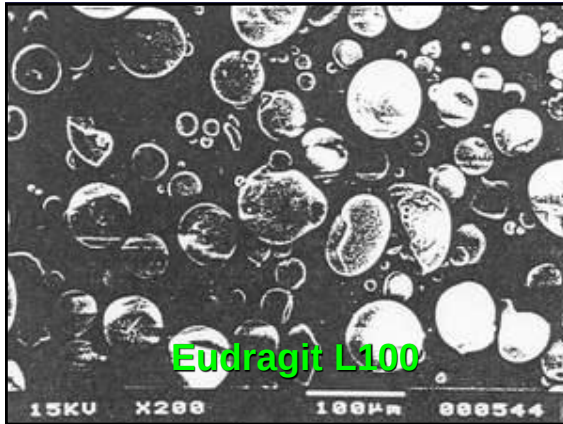
---

---

---

---

---




---

---

---

---

---

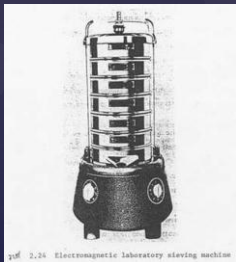
---

---

---

### 3.1.2 Sieving

- Quantitative particle size distribution analysis.
- For size > 50 µm upward.
- Shape has strong influence on results.



รูป 2.26 Electromagnetic laboratory sieving machine

---

---

---

---

---

---

---

---

ตารางที่ 2.4 ขนาดของหน้าผัด U.S. Standard และ Tyler Standard

| U.S. Standard |      |        |      | Tyler Standard |      |        |      |
|---------------|------|--------|------|----------------|------|--------|------|
| Micron        | Mesh | Micron | Mesh | Micron         | Mesh | Micron | Mesh |
| 5660          | 3½   | 420    | 40   | 5613           | 3½   | 351    | 42   |
| 4760          | 4    | 350    | 45   | 4699           | 4    | 295    | 48   |
| 4000          | 5    | 297    | 50   | 3965           | 5    | 246    | 60   |
| 3360          | 6    | 250    | 60   | 3327           | 6    | 208    | 65   |
| 2830          | 7    | 210    | 70   | 2794           | 7    | 175    | 80   |
| 2380          | 8    | 177    | 80   | 2362           | 8    | 147    | 100  |
| 2000          | 10   | 149    | 100  | 1651           | 10   | 124    | 115  |
| 1690          | 12   | 125    | 120  | 1397           | 12   | 104    | 150  |
| 1410          | 14   | 105    | 140  | 1168           | 14   | 88     | 170  |
| 1190          | 16   | 88     | 170  | 991            | 16   | 74     | 200  |
| 1000          | 18   | 74     | 200  | 883            | 20   |        |      |
| 840           | 20   | 62     | 230  | 701            | 24   |        |      |
| 710           | 25   | 53     | 270  | 589            | 28   |        |      |
| 590           | 30   | 44     | 325  | 495            | 32   |        |      |
| 500           | 35   | 37     | 400  | 417            | 35   |        |      |

---

---

---

---

---

---

---

---

ตารางที่ 2.5 Size-weight distribution ของ sodium bromide เกณฑ์  
กำหนดโดย U.S. Standard Sieves

| Sieve Number<br>(Passed/Retained) | Arithmetic Mean<br>Size of<br>Openings | Weight<br>Retained on<br>Smaller Sieve | % Retained<br>on Smaller<br>Sieve | Weight<br>Size |
|-----------------------------------|----------------------------------------|----------------------------------------|-----------------------------------|----------------|
| (1)                               | (2)                                    | (3)                                    | (4)                               | (2) x (4)      |
| 30/45                             | 470 $\mu$                              | 57.3 Gm.                               | 13.0                              | 6100           |
| 45/60                             | 300                                    | 181.0                                  | 41.2                              | 12,380         |
| 60/80                             | 213                                    | 110.0                                  | 25.0                              | 5320           |
| 80/100                            | 163                                    | 49.7                                   | 11.3                              | 1840           |
| 100/140                           | 127                                    | 20.0                                   | 4.5                               | 572            |
| 140/200                           | 90                                     | 22.0                                   | 5.0                               | 450            |
|                                   |                                        | 440.0                                  | 100.0                             | 26,662         |

$d_{ave} = \frac{26,662}{100} = 267 \mu$

### 3.1.3 Electronic means

To encompass most pharmaceutical  
powders ranging in size 1 - 120  $\mu$ m:

- Blockage of electrical conductivity path (Coulter)
- Light blockage (HIAC) [adopted by USP]
- Light scattering (Royco)
- Laser scattering (Malvern)

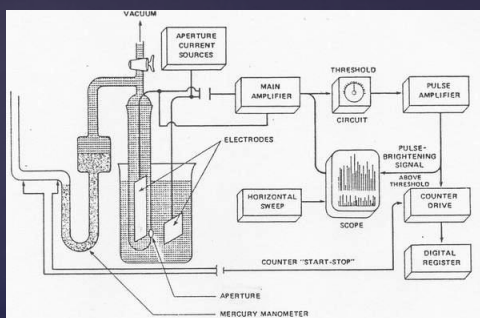
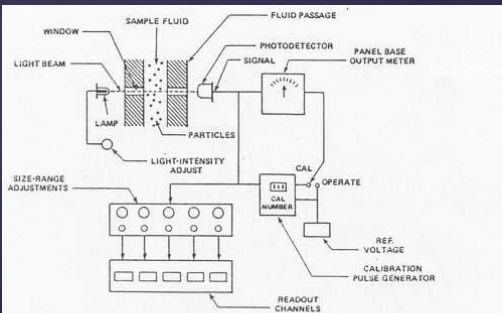


Figure 55. Schematic of the Coulter Counter. (Courtesy Coulter Electronics, Inc., Hialeah, Florida 33010.)

Electrical conductivity method



Light obstruction method

Hine/Royco

Summary Report

Sample ID1: C114b Inj. (nupul)

Sample ID2: not analyzed

Sample volume/ml: 10.0

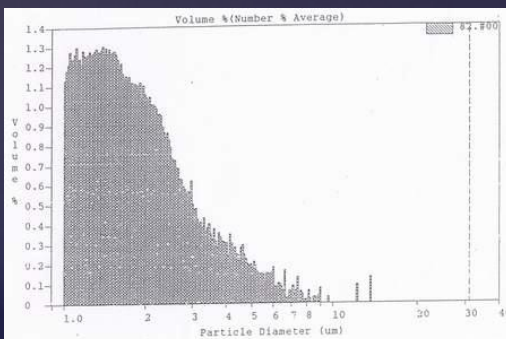
Sample date: 24/06/88

Sensor model: HRI5-150

Sensor serial: 95020063

Operator name: Narpang

| Chan   | Size<br>um | Count<br>Counts | Diff<br>Counts | Count<br>Counts/ml | Diff<br>Counts/ml |
|--------|------------|-----------------|----------------|--------------------|-------------------|
| Run: 1 |            |                 |                |                    |                   |
| 01     | 2.50       | 12006           | 7871           | 1200.6             | 787.1             |
| 02     | 5.00       | 4135            | 2781           | 413.5              | 278.1             |
| 03     | 7.50       | 1254            | 755            | 125.4              | 75.5              |
| 04     | 10.00      | 290             | 116            | 29.0               | 11.6              |
| 05     | 12.50      | 93              | 17             | 9.3                | 1.7               |
| 06     | 15.00      | 136             | 17             | 13.6               | 1.7               |
| 07     | 17.50      | 89              | 29             | 8.9                | 2.9               |
| 08     | 20.00      | 60              | 22             | 6.0                | 2.2               |
| 09     | 22.50      | 38              | 12             | 3.8                | 1.2               |
| 10     | 25.00      | 26              | 8              | 2.6                | 0.8               |
| 11     | 27.50      | 17              | 3              | 1.7                | 0.3               |
| 12     | 30.00      | 14              | 2              | 1.4                | 0.2               |
| 13     | 32.50      | 12              | 3              | 1.2                | 0.3               |
| 14     | 35.00      | 9               | 2              | 0.9                | 0.2               |
| 15     | 37.50      | 7               | 2              | 0.7                | 0.2               |
| 16     | 40.00      | 6               | 5              | 0.6                | 0.5               |



## 3.1.4 Other techniques

- Centrifugation
- Air suspension
- Sedimentation (Adreassen pipet, recording balance)

Disfavor now because of their tedious nature.

Table 3. Common Techniques for Measuring Fine Particles of Various Sizes

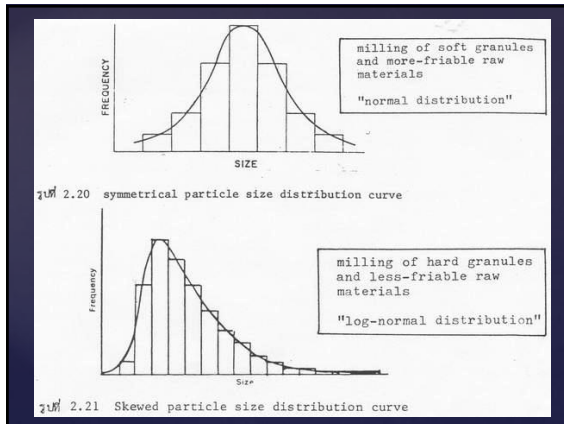
| Technique        | Particle size ( $\mu\text{m}$ ) |
|------------------|---------------------------------|
| Microscopic      | 1 - 100                         |
| Sieve            | > 50                            |
| Sedimentation    | > 1                             |
| Elutriation      | 1 - 50                          |
| Centrifugal      | < 50                            |
| Permeability     | > 1                             |
| Light scattering | 0.5 - 50                        |

(Parrott)

Table 16  
Common Particle Counting Methods and Resulting Average Particle Size

| Method of counting                | Approximate size range covered                              | Resulting average particle size                                                                                               |
|-----------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1. Optical microscope             | 0.5-100 $\mu\text{m}$<br>(140 mesh)                         | Average by number, $\bar{N}_n$<br>Average by surface, $\bar{N}_s$ <sup>a</sup><br>Average by volume, $\bar{N}_v$ <sup>a</sup> |
| 2. Sedimentation in gas or liquid | 2-200 $\mu\text{m}$<br>(70 mesh)                            | Average by weight, $\bar{N}_w$                                                                                                |
| 3. Electronic sensing zone        | 1-300 $\mu\text{m}$<br>(50 mesh)                            | Average by volume, $\bar{N}_v$<br>(volume of equivalent sphere)                                                               |
| 4. Light obstruction              | 2-300 $\mu\text{m}$<br>(50 mesh)                            | Average by volume, $\bar{N}_v$ ,<br>as calculated from cross-sectional area of an equivalent sphere                           |
| 5. Sieve                          | 44 $\mu\text{m}$ (325 mesh)-<br>4700 $\mu\text{m}$ (2 mesh) | Average by weight, $\bar{N}_w$                                                                                                |
| 6. Air permeation                 | 0.05-150 $\mu\text{m}$<br>(100 mesh)                        | Average by surface and volume, $\bar{N}_{vs}$                                                                                 |

<sup>a</sup>Per calculations shown from number distribution.




---

---

---

---

---

---

---

---

---

---

Table 4 Particle Size Distribution of a Ground Sample of Triamcinolone Acetonide

| Size range (μm) | No. of particles | Volume of particles $\times 10^{-3}$ (μm <sup>3</sup> ) | Weight percent in range | (Undersize) Cumulative weight percent |
|-----------------|------------------|---------------------------------------------------------|-------------------------|---------------------------------------|
| 22.5–26.5       | 5                | 38                                                      | 0.2                     | 100.0                                 |
| 18.6–22.0       | 54               | 237                                                     | 1.7                     | 99.8                                  |
| 14.9–18.6       | 488              | 1212                                                    | 8.8                     | 98.1                                  |
| 11.8–14.9       | 2072             | 2552                                                    | 18.5                    | 89.3                                  |
| 9.4–11.8        | 5376             | 3352                                                    | 24.3                    | 70.8                                  |
| 7.4–9.4         | 9632             | 2989                                                    | 21.7                    | 46.5                                  |
| 5.9–7.4         | 12,544           | 1888                                                    | 13.7                    | 24.8                                  |
| 4.7–5.9         | 12,928           | 1008                                                    | 7.3                     | 11.1                                  |
| 3.7–4.7         | 13,568           | 526                                                     | 3.8                     | 3.8                                   |

---

---

---

---

---

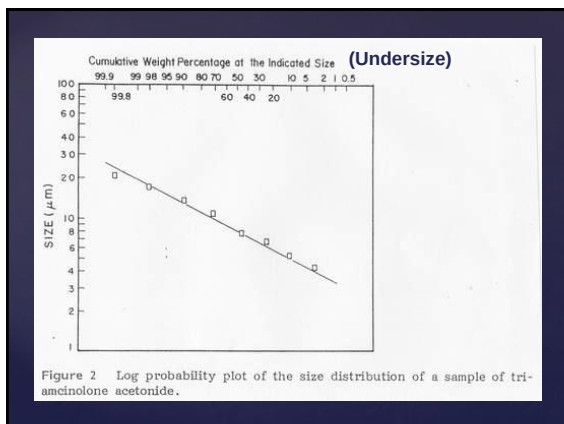
---

---

---

---

---




---

---

---

---

---

---

---

---

---

---

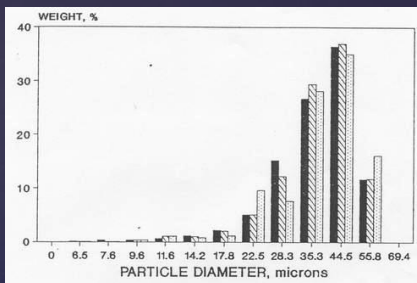
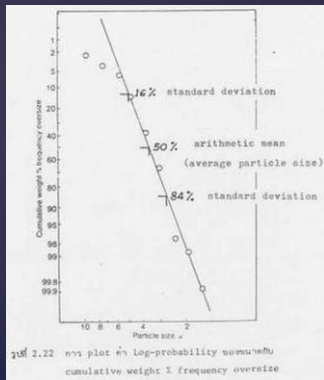


Figure 57. The histogram plots of particle diameter in percentage weight as a function of particle diameter for sulfamethoxazole.

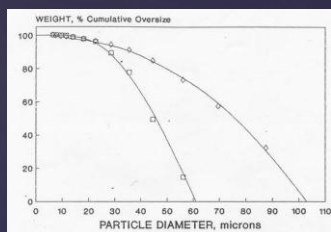


Figure 56. Distribution of particle sizes for sulfamethoxazole and trimethoprim obtained with the Coulter counter. Smooth curves of sulfamethoxazole and trimethoprim are in accordance with  $n = \bar{x}_{sulf} / (\Delta x_s)$  and  $n = \bar{x}_{thp} / (\Delta x_t)$ , respectively.

Key : ○, SMX; □, THP.



### 3.2 Determination of Surface Area

#### ↳ Surface areas of powders

-> increasing attention in recent years: reflect the particle size

#### ↳ Grinding operation:

↓ particle size ==> ↑ surface area.

#### ↳ Brunauer-Emmett-Teller (BET) theory of adsorption

Most substances will adsorb a monomolecular layer of a gas under certain conditions of partial pressure (of the gas) and temperature.

Knowing the monolayer capacity of an adsorbent (i.e., the quantity of adsorbate that can be accommodated as a monolayer on the surface of a solid, the adsorbent) and the area of the adsorbate molecule, the surface area can, in principle be calculated.

Most commonly, nitrogen is used as the adsorbate at a specific partial pressure established by mixing it with an inert gas, typically helium. The adsorption process is carried out at liquid nitrogen temperature (-195 °C).

It has been demonstrated that, at a partial pressure of nitrogen attainable when it is in a 30 % mixture with an inert gas and at -195 °C, a monolayer is adsorbed onto most solids.

Apparently, under these conditions the polarity of nitrogen is sufficient for van de Waals forces of attraction between the adsorbate and the adsorbents to be manifest.

The kinetic energy present under these conditions overwhelms the intermolecular attraction between nitrogen atoms. However, it is not sufficient to break the bonding between the nitrogen and dissimilar atoms. The latter are most often more polar and prone to van de Waals forces of attraction.

The nitrogen molecule does not readily enter into chemical combinations, and thus its binding is of a nonspecific nature (i.e., it enters into a physical adsorption); consequently, the nitrogen molecule is well suited for this role.

### Brunauer-Emmett-Teller (BET) adsorption isotherm

$$\frac{1}{\lambda(P_o/P - 1)} = \frac{C - 1}{\lambda_m C} \frac{P}{P_o} + \frac{1}{\lambda_m C} \quad (1)$$

$\lambda$  = g of adsorbate per g of adsorbent

$\lambda_m$  = maximum value of that  $\lambda$  ratio for a monolayer

$P$  = partial pressure of the adsorbate gas

$P_o$  = vapor pressure of the pure adsorbate gas

$C$  = constant

$P_o$ , and  $C$  are temperature-dependent

- ↳ The values of  $\lambda$  (g of adsorbate/g of adsorbent) at various  $P$  values (partial pressure of the adsorbate gas) could be obtained from the experiment through instrument.
- ↳  $P_0$  (vapor pressure of the pure adsorbate gas) can be obtained from the literature.
- ↳ Plotting the term  $1/[\lambda(P_0/P - 1)]$  against  $P/P_0$  will obtain a straight line with
 
$$\text{slope} = (C - 1)/\lambda_m C$$

$$\text{intercept} = 1/\lambda_m C$$
- ↳ The term  $C$  and  $\lambda_m$  can readily be obtained

---

---

---

---

---

---

---

---

### Dynamic Method of Gas Adsorption

- ↳ Accurately weighing the sample into an appropriate container
- ↳ Immersing the container in liquid nitrogen
- ↳ Passing the gas over the sample
- ↳ Removing the liquid nitrogen when the adsorption is complete (as signaled by the instrument)
- ↳ Warming the sample to about the room temperature
- ↳ Measuring (via the instrument) the adsorbed gas released (column 3 of Table 5)
- ↳ Performing the calibration by injecting known amounts of adsorbed gas into the proper instrument port (column 4 and 5 of Table 5)
- ↳  $P$  is the product of the fraction of  $N_2$  in the gas mixture (column 1 of Table 5) and the ambient pressure

---

---

---

---

---

---

---

---

Table 5 Specific Surface Area of a Sample of Sodium Epicillin<sup>a</sup>

| $N_2$<br>in<br>He | $P/P_0$ | Standardization  |                           |                  |                                              | $[\lambda(P_0/P - 1)]^{-1}$ |
|-------------------|---------|------------------|---------------------------|------------------|----------------------------------------------|-----------------------------|
|                   |         | Signal<br>(area) | Vol $N_2$<br>used<br>(ml) | Signal<br>(area) | Wt $N_2$<br>absorbed<br>$\times 10^{-4}$ (g) |                             |
| 4.9               | 0.0483  | 178              | 0.050                     | 137              | 0.576                                        | 677.5                       |
| 9.7               | 0.0956  | 152              | 0.080                     | 150              | 0.922                                        | 1130.9                      |
| 20.0              | 0.1972  | 248              | 0.100                     | 238              | 1.153                                        | 2043.6                      |
| 29.9              | 0.2948  | 312              | 0.130                     | 331              | 1.499                                        | 2957.5                      |

<sup>a</sup> Sample weight: 0.1244 g. Atmospheric pressure: 762.7 mmHg. Temperature: 297 K. Results: Slope, 9210; y Intercept, 238.2; correlation coefficient, 0.99995; specific surface area,  $3.0 \text{ m}^2 \text{ g}^{-1}$ .

---

---

---

---

---

---

---

---

Table 6 Relationship Between Diameter of a Particle and Specific Surface Area

| Diameter ( $\mu\text{m}$ ) | Specific surface area ( $\text{m}^2 \text{g}^{-1}$ ) |
|----------------------------|------------------------------------------------------|
| 0.25                       | 24                                                   |
| 0.50                       | 12                                                   |
| 1.0                        | 6                                                    |
| 2.0                        | 3                                                    |
| 4.0                        | 1.5                                                  |
| 10.0                       | 0.63                                                 |
| 15.0                       | 0.4                                                  |
| 20.0                       | 0.3                                                  |
| 40.0                       | 0.15                                                 |

---

---

---

---

---

---

---

---

- At relatively large diameters, the specific surface area is insensitive to an increase in diameter
- At very small diameters the surface area is comparatively very sensitive.
- Relatively high surface area most often reflects a relatively small particle size, except porous or strongly agglomerated mass
- Small particles (thus of high surface area) agglomerate more readily, and often to render the inner pores and surfaces inaccessible to dissolution medium

---

---

---

---

---

---

---

---

#### 4. SOLUBILITY

- Solubility  $> 1\%$  w/v  
=> no dissolution-related absorption problem
- Highly insoluble drug administered in small doses may exhibit good absorption
- Unstable drug in highly acidic environment of stomach, high solubility and consequent rapid dissolution could result in a decreased bioavailability
- The solubility of every new drug must be determined as a function of pH over the physiological pH range of 1 - 8

---

---

---

---

---

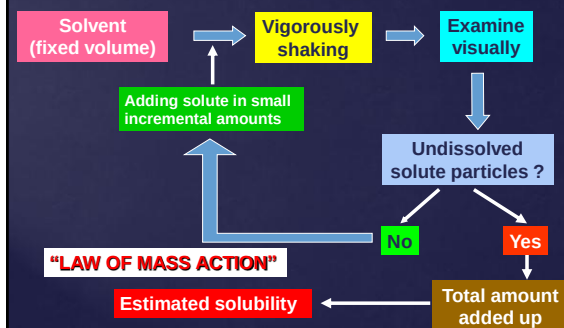
---

---

---

## 4.1 Determination of Solubility

### 4.1.1 Semiquantitative determination:




---

---

---

---

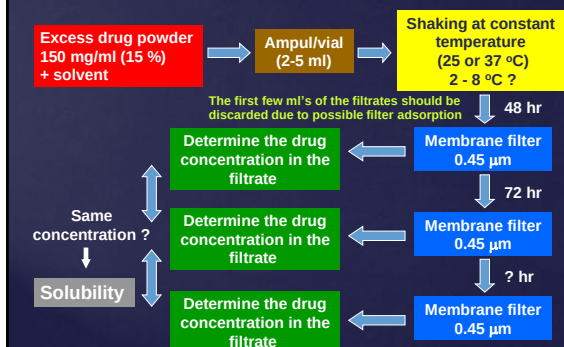
---

---

---

---

### 4.1.2 Accurately Quantitative determination:




---

---

---

---

---

---

---

---

### 4.1.3 Unique Problems in Solubility Determination of Poorly Soluble Compounds

- ⌘ Solubilities could be overestimated due to the presence of soluble impurities
- ⌘ Saturation solubility is not reached in a reasonable length of time unless the amount of solid used is greatly in excess of that needed to saturation
- ⌘ Many compounds in solution degrade, thus making an accurate determination of solubility difficult
- ⌘ Difficulty is also encountered in the determination of solubility of metastable forms that transform to more stable forms when exposed to solvents

---

---

---

---

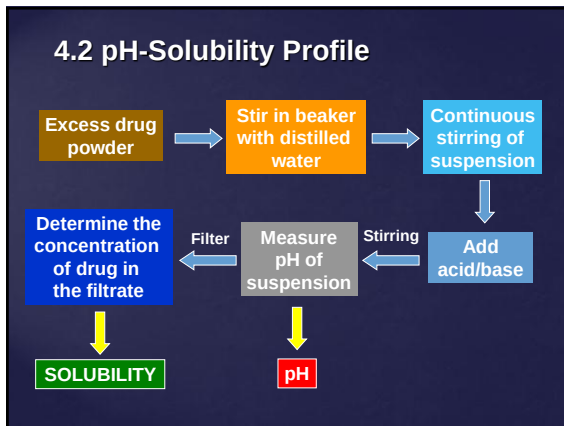
---

---

---

---

## 4.2 pH-Solubility Profile




---

---

---

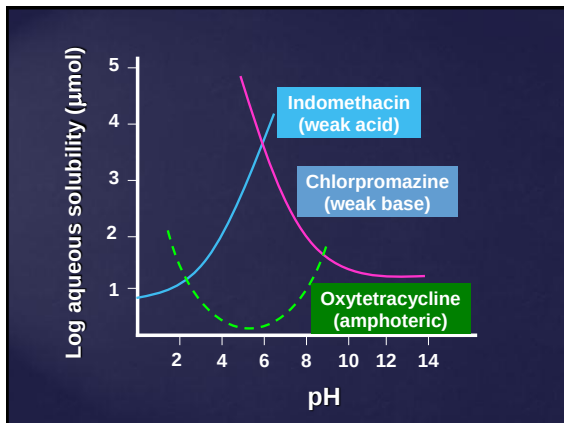
---

---

---

---

---




---

---

---

---

---

---

---

---

↳ Poorly-soluble weakly-acidic drugs:

$$\text{pH} = \text{pK}_a + \log \left[ \frac{S_t - S_o}{S_o} \right] \quad (2)$$

↳ Poorly-soluble weakly-basic drugs:

$$\text{pH} = \text{pK}_a + \log \left[ \frac{S_o}{S_t - S_o} \right] \quad (3)$$

where

$S_o$  = solubility of unionized free acid or base

$S_t$  = total solubility (unionized + ionized)

---

---

---

---

---

---

---

---

### 4.3 Salt Forms

NSAID's      alclofenac, diclofenac, fenbufen,  
                  ibuprofen, naproxen

Weak acid  $pK_a \sim 4$ , low solubility

Salt forms      sodium

N-(2-hydroxy ethyl) piperazinium  
arginium

N-methylglucosammonium

Solubility

diclofenac (free acid) :       $0.8 \times 10^{-5}$  M (25 °C)

diclofenac sodium :      24.5 mg/ml (37 °C)

### 4.3 Salt Forms (cont.)

Quinolones      enoxacin, norfloxacin,  
                      ciprofloxacin

Salt forms      lactate, acetate, gluconate,  
                      galacturonate, aspartate,  
                      glutamate, etc.

Solubility

Free base :      < 0.1 mg/ml (25 °C)

Salt forms :      > 100 mg/ml (25 °C)

| Combining Compound    | (Enoxacin)     |                     |                      | (Norfloxacin)       |                      |                      |
|-----------------------|----------------|---------------------|----------------------|---------------------|----------------------|----------------------|
|                       | Compound 1     | Compound 2          | Compound 3           | Compound 1          | Compound 2           | Compound 3           |
| pK <sub>a</sub> (9)   | F. Wt.         | Solubility<br>mg/ml | pH of<br>Supernatant | Solubility<br>mg/ml | pH of<br>Supernatant | pH of<br>Supernatant |
| <b>Acids</b>          |                |                     |                      |                     |                      |                      |
| Ascorbic              | 4.76           | 195.05              | 5.0                  | 235                 | 5.3                  |                      |
| Citric                | 3.1, 4.8, 6.4  | 210.14              | 1.3                  | 1.3                 | 3.3                  |                      |
| Galacturonic          | 2.8            | 196.14              | 155                  | 4.8                 | 166                  | 5.4                  |
| Gluconic              | 3.2            | 196.16              | 155                  | 4.6                 | 162                  | 4.3                  |
| Hydrochloric          | 7              | +                   | 1.0                  | 201                 | 1.5                  |                      |
| Isohydric             | +              | 220                 | 1.0                  | 175                 | 1.4                  |                      |
| Lactonic              | 2.5            | 358.3               | 99                   | 46.5                | 4.7                  |                      |
| Lactulose             | 2.5, 6.2       | 90.8                | 4.2                  | 199                 | 4.1                  |                      |
| Lactide               | 3.9            | 217                 | 2.6                  | 45.1                | 2.8                  |                      |
| Maleic                | 2.5, 6.2       | 114.07              | 1.7                  | 175                 | 1.3                  |                      |
| Malic                 | 1.3, 4.5       | 126.07              | 3.8                  | 19.8                | 2.1                  |                      |
| Malononitrile         | 4.2, 5.6       | 118.09              | 7.5                  | 128                 | 4.3                  |                      |
| Tartaric              | 3.0, 4.4       | 150.09              | 3.3                  | 3.5                 | 3.0                  |                      |
| <b>Amino acids</b>    |                |                     |                      |                     |                      |                      |
| L-Arginine            | 2.2, 9.1, 13.2 | 174.2               | 6.7                  | 9.7                 | 11.1                 | 9.8                  |
| L-Aspartic            | 2.0, 3.2, 10.1 | 123.1               | 165                  | 4.6                 | 174                  |                      |
| L-Cysteine            | 1.9, 8.7       | 169.16              | 221                  | 2.8                 | 174                  | 2.9                  |
| L-Glutamic            | 2.2, 4.3, 9.7  | 147.13              | 160                  | 5.0                 | 174                  |                      |
| Glycine               | 2.3            | 75.07               | 1.2                  | 6.9                 | 9.9                  | 6.1                  |
| L-Isoleucine          | 2.3, 8.0       | 131.2               | 1.3                  | 6.9                 | 6.2                  |                      |
| L-Leucine             | 2.3, 9.1, 10.8 | 146.2               | 4.3                  | 9.4                 | 4.2                  | 9.5                  |
| L-Serine              | 2.2, 4.1       | 105.09              | 1.3                  | 6.7                 | 1.0                  | 6.2                  |
| <b>Amino alcohols</b> |                |                     |                      |                     |                      |                      |
| Choline               | 121.18         | 38.2                | 9.8                  | 53.5                | 10.3                 |                      |
| Diethanolamine        | 105.07         | 4.3                 | 9.7                  | 7.5                 | 9.7                  |                      |
| NaOH <sup>a</sup>     | +              | 75                  | 10.5                 | 131                 | 10.0                 |                      |

<sup>a</sup>Reference at selected solubility once made from concentrate.

<sup>b</sup>Back response precipitated when solutions were adjusted to pH 7.4.

ciprofloxacin lactate for injection  
(Ciprobay<sup>®</sup> Infusion)

## 4.4 Solubilization

Drug not an acidic or basic, or the acidic or basic character not amenable to the formation of a stable salt

- ↳ Use more soluble metastable polymorph
- ↳ Use of complexation (eg. Riboflavin-xanthines complex)
- ↳ Use of high-energy coprecipitates that are mixtures of solid solutions and solid dispersions (eg. Griseofulvin in PEG 4000, 6000, and 20,000)
  - in PEG 4000 and 20,000 -> supersaturated solutions
  - in PEG 6000 -> bioavailability in human twice > micronized drug
- ↳ Use of suitable surfactant

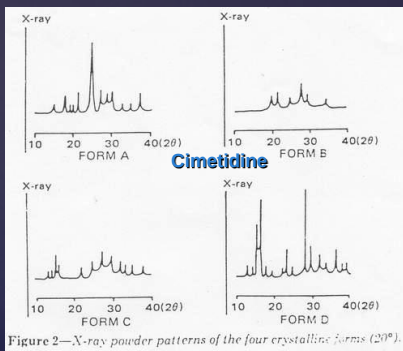
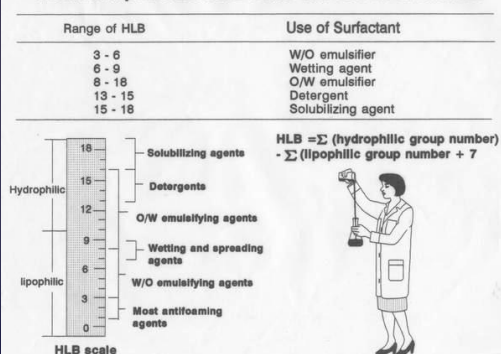


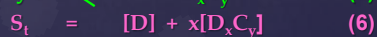
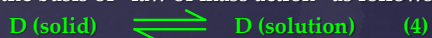
Figure 2—X-ray powder patterns of the four crystalline forms (20°).

### Relationship between HLB Value and Use of Surfactants



### 4.4.1 Complexation

Complexation can be analyzed and explained on the basis of "law of mass action" as follows:

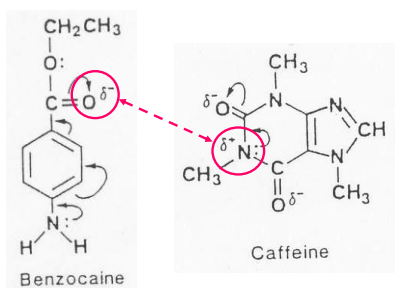


where

D = drug molecule

C = complexing agent (ligand)

$S_t$  = total solubility of free drug [D] and the drug in the complex  $[D_xC_y]$



### Ligand (Complexing Agents)

- Vitamin K
- Menadione
- Cholesterol
- Cholate salt
- $\beta$ -cyclodextrin
- Caffeine
- Benzoic acid
- PEG series
- PVP

*Formulation point of view:*

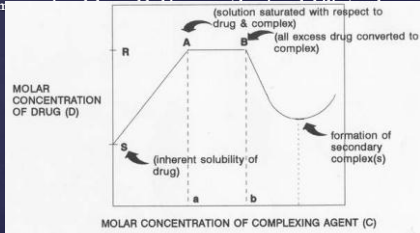
1. How much will a specific complexing agent be used for a certain amount of drug?
2. How does the resultant complex affect the safety, stability, and therapeutic efficacy of the product?



Stoichiometric ratio =  $\frac{\text{moles of drug in complex}}{\text{moles of complexing agent in the complex}}$  (7)

$$x:y = \frac{D_T - R}{b - a} \quad (8)$$

$D_T = \text{Amount of drug added}$



Phase diagram showing the effect of varying concentrations of complexing agent on the apparent solubility of drug

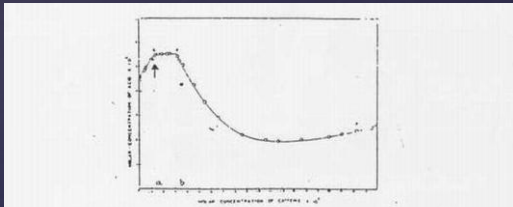


Fig. 14 Phase diagram of PABA-caffeine system in water at 30°C

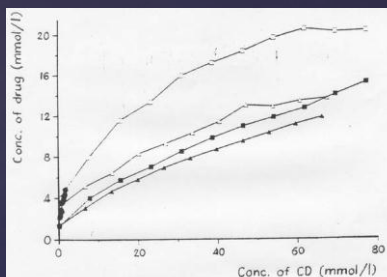


Fig. 1. Phase solubility diagram of indomethacin and piroxicam in phosphate buffer pH 7.4 at 25°C. (□) Indomethacin-HP-β-CD MS 0.39; (Δ) indomethacin-HP-γ-CD MS 0.4; (■) piroxicam-HP-β-CD MS 0.39; (▲) piroxicam-HP-γ-CD MS 0.4; (●) piroxicam + β-CD.

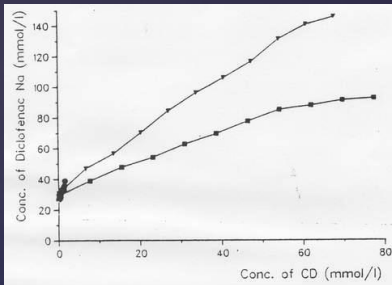


Fig. 2. Phase solubility diagram of diclofenac Na in phosphate buffer pH 7.4 at 25 °C. (■) Diclofenac Na+HP-β-CD MS 0.39; (▼) diclofenac Na+HP-γ-CD MS 0.4; (●) diclofenac Na+β-CD.

## 5. Dissolution

$k_d \ll k_a \Rightarrow$  "dissolution rate-limited"

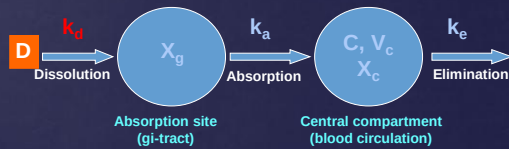
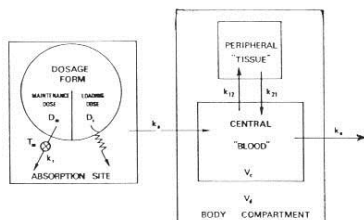
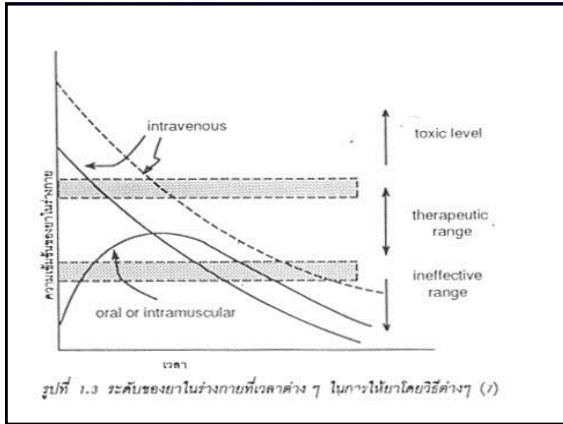


Diagram showing dissolution and absorption of solid dosage form into blood circulation



รูปที่ 4.4 แผนภาพโมเดลทางเภสัชจลนศาสตร์ (pharmacokinetic model) ที่ใช้ ในทางทฤษฎีของระบบเภสัชจลนศาสตร์ (2)



## 5.1 INTRINSIC DISSOLUTION

### 5.1.1 Film Theory

The dissolution of a solid in its own solution is adequately described by **Noyes-Nernst's "Film Theory"**

$$-\frac{dW}{dt} = \frac{DAK}{h} (C_s - C) \quad (9)$$

where

$dW/dt$  = dissolution rate

$A$  = surface area of the dissolving solid

$D$  = diffusion coefficient

$K$  = partition coefficient

$h$  = aqueous diffusion layer

$C_s$  = solubility of solute

$C$  = solute concentration in the bulk medium

## 5.1 Intrinsic Dissolution

### 5.1.1 Film Theory

The dissolution of a solid in its own solution is adequately described by **Noyes-Nernst's "Film Theory"**

$$-dW/dt = ADK(C_s - C)/h$$

$dW/dt$  = dissolution rate of solid

$A$  = surface area of dissolving solid

$D$  = diffusion coefficient

$K$  = partition coefficient

$C_s$  = solubility of solute

$C$  = solute concentration in bulk medium

$h$  = aqueous diffusion layer thickness



↳ **Intrinsic dissolution rate ( $\text{mg}/\text{cm}^2/\text{min}$ )** is characteristics of each solid compound in a given solvent under fixed hydrodynamic conditions

↳ **Intrinsic dissolution rate** helps in predicting if absorption would be **dissolution rate-limited**

↳  $> 1 \text{ mg}/\text{cm}^2/\text{min}$  --> not likely to present dissolution rate-limited absorption problems

↳  $< 0.1 \text{ mg}/\text{cm}^2/\text{min}$  --> usually exhibit dissolution rate-limited absorption

↳  $0.1 - 1.0 \text{ mg}/\text{cm}^2/\text{min}$  --> more information is needed before making any prediction

---

---

---

---

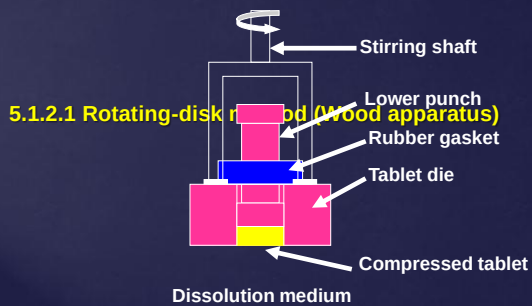
---

---

---

---

### 5.1.2 Method of Determination




---

---

---

---

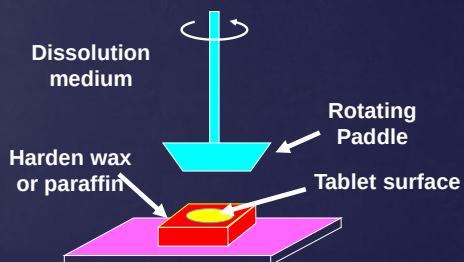
---

---

---

---

### 5.1.2.2 Nelson Constant Surface Method




---

---

---

---

---

---

---

---

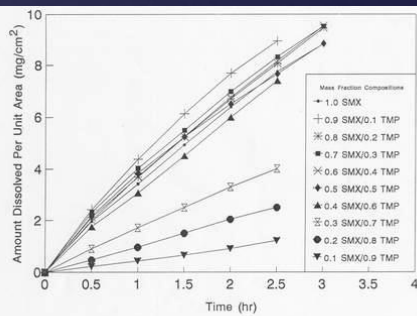


Figure 1. Dissolution profiles of sulfamethoxazole from sulfamethoxazole (SMX) - trimethoprim (TMP) solid mixtures with various mass fraction compositions.

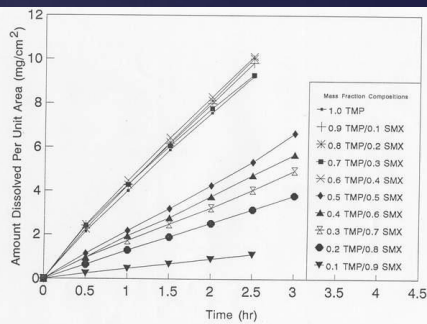
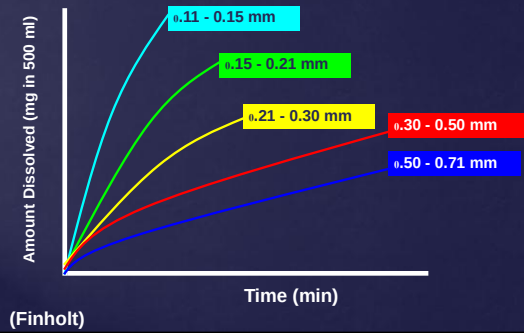


Figure 2. Dissolution profiles of trimethoprim from sulfamethoxazole (SMX) - trimethoprim (TMP) solid mixtures with various mass fraction compositions.

## 5.2 Particulate Dissolution

- ↳ Particulate dissolution is used to study the influence on dissolution of particle size, surface area, and mixing with excipients.
- ↳ The rate of dissolution normally increased with a decrease in the particle size.
- ↳ Occasionally, however, an inverse relationship of particle size to dissolution is encountered.
- ↳ This may be explained on the basis of effective or available, rather than absolute, surface area; and it is caused by incomplete wetting of the powder.
- ↳ Incorporation of a surfactant in the dissolution medium may provide the expected relationship.

### 5.2.1 Effect of particle size of phenacetin on dissolution rate of the drug from granules




---

---

---

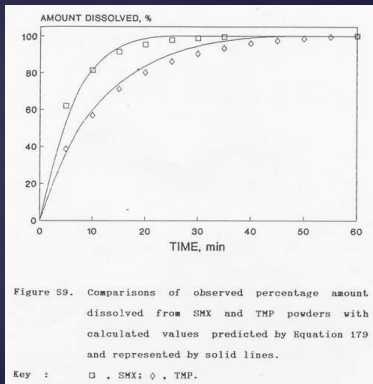
---

---

---

---

---




---

---

---

---

---

---

---

---

### 5.2.2 Means of enhancing the slow dissolution:

in absence of more soluble physical or chemical form of the drug -

- ↳ Particle size reduction (most commonly used).
- ↳ Enhanced surface area by adsorbing the drug on an inert excipient with a high surface area, i.e., fumed silicon dioxide.
- ↳ Comelting, coprecipitating, or triturating the drug with some excipients.
- ↳ Incorporation of suitable surfactant.

---

---

---

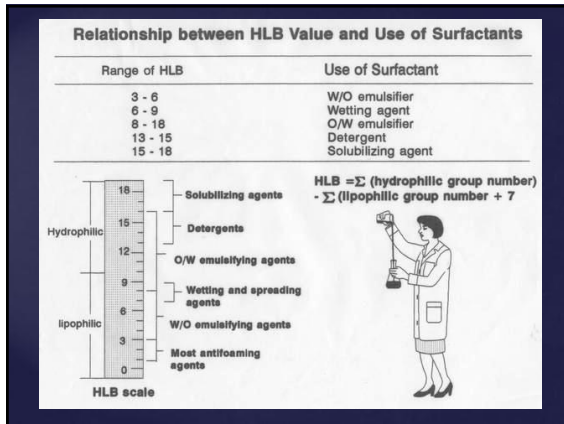
---

---

---

---

---



### 5.3 Prediction of Dissolution Rate

Consider the dissolution of **22 mg** of **60/80 mesh hydrocortisone** in **500 ml** of water. The aqueous solubility of hydrocortisone is **0.28 mg/ml**. The 60/80 mesh fraction corresponds to **212  $\mu\text{m}$**  or  **$2.12 \times 10^{-2}$  cm** in diameter. The density of hydrocortisone is **1.25 g/ml**. The volume of a sphere is  **$(4/3)\pi r^3$** . Assuming that all particles are spheres of the same diameter, 22 mg would correspond to

$$\frac{22 \times 10^{-3}}{1.25 \frac{4\pi}{3} \times (1.06)^3 \times 10^{-6}} = 3,500 \text{ spherical particles}$$

The area of a sphere is given by  $4\pi r^2$ . Therefore, the area of 3,500 particles of average radius  $1.06 \times 10^{-2}$  cm is

$$4\pi \times (1.06)^2 \times 10^{-4} \times 3,500 = 4.94 \text{ cm}^2$$

The dissolution rate according to Eq.(9) is

$$\frac{-dW}{dt} = \frac{DAK}{h} (C_s - C) \quad (9)$$

where

$D = 9.0 \times 10^{-6} \text{ cm}^2/\text{sec}$  (good approximation for most drugs)

$A = 4.94 \text{ cm}^2$

$K = 1.0$

$h = 5.0 \times 10^{-3} \text{ cm}$  (diffusion layer thickness at 50 rpm stirring)

$C_s = 0.28 \text{ mg/ml}$

$C = 0$  (early phase of dissolution)

Thus, for the sample of hydrocortisone,

$$\begin{aligned} \text{Initial dissolution rate} &= \frac{4.94 \times 9.0 \times 10^{-6} \times 0.28}{5.0 \times 10^{-3}} \\ &= 2.49 \times 10^{-3} \text{ mg/sec} \end{aligned}$$

## 6. Parameter Affecting Absorption

The absorption of drugs administered orally as solids consists of 2 consecutive processes:

1. The process of dissolution, followed by
2. The transport of the dissolved materials across gi membranes into systemic circulation

---

---

---

---

---

---

---

The rate-determining step in the overall absorption process:

**For relatively insoluble compounds**

-> rate of dissolution

(can be altered via physical intervention)

**For relatively soluble compounds**

-> rate of permeation across biological membrane

(is dependent on size, relative aqueous and lipid solubilities, and ionic charge of the solute molecules)

(can be altered, in the majority of cases, only through molecular modification)

---

---

---

---

---

---

---

↳ In making a judgement concerning the absorption potential of a new drug entity, the preformulation scientist must undertake studies to delineate its dissolution as well as permeation behavior.

↳ Characterization of the permeation behavior of a new drug must be performed at an early stage of drug development-primarily to help avoid mistaken efforts to improve its absorption by improving dissolution, when in reality the absorption is permeability-limited.

↳ **PERMEABILITY STUDIES ARE OF EVEN GREATER IMPORTANCE WHEN ANALOGS OF THE COMPOUND HAVING SIMILAR PHARMACOLOGICAL ATTRIBUTES ARE AVAILABLE**

↳ Permeability studies then would aid in the selection of the compound with the greatest absorption potential.

---

---

---

---

---

---

---



## 6.1 Partition Coefficient

- ↳ Like biological membrane in general, the gi membranes are largely lipoidal in character.
- ↳ The rate and extent of absorption decreased with the increasing polarity of molecules.
- ↳ **Partition coefficient (distribution coefficient):** the ratio in which a solute distributes itself between the two phases of two immiscible liquids that are in contact with each other (mostly n-octanol/water).

---

---

---

---

---

---

---

---

Comparison Between Colonic Absorption and Lipid/Water Partition of the Un-ionized forms of Barbiturates

| Barbiturate          | % Absorbed | Chloroform/water partition coefficient |  |
|----------------------|------------|----------------------------------------|--|
| Barbital             | 12 ± 2     | 0.7                                    |  |
| Aprobarbital         | 17 ± 2     | 4.0                                    |  |
| Phenobarbital        | 20 ± 3     | 4.8                                    |  |
| Allylbarbituric acid | 23 ± 3     | 10.5                                   |  |
| Butethal             | 24 ± 3     | 11.7                                   |  |
| Cyclobarbital        | 24 ± 3     | 18.0                                   |  |
| Pentobarbital        | 30 ± 2     | 23.0                                   |  |
| Secobarbital         | 40 ± 3     | 50.7                                   |  |
| Hexethal             | 44 ± 3     | > 100.0                                |  |
| (Schanker)           |            |                                        |  |

---

---

---

---

---

---

---

---

## 6.2 Ionization Constant

- ↳ The unionized species are more lipid-soluble and hence more readily absorbed.
- ↳ The gi absorption of weakly acidic or basic drugs is related to the fraction of unionized drug in solution.
- ↳ **Factors affecting absorption:**
  - pH at the site of absorption
  - Ionization constant
  - Lipid solubility of unionized species

“pH-partition theory”

---

---

---

---

---

---

---

---

**Henderson-Hasselbalch equation****For acids:**

$$\text{pH} = \text{pK}_a + \log \frac{[\text{ionized form}]}{[\text{unionized form}]}$$

**For bases:**

$$\text{pH} = \text{pK}_a + \log \frac{[\text{unionized form}]}{[\text{ionized form}]}$$

**Determination of Ionization Constant**

1. Potentiometric pH-Titration
2. pH-Spectrophotometry Method
3. pH-Solubility Analysis

---

---

---

---

---

---

---